with ˆy the income of (15). A holder keeps its voucher while it is eligible, and an eligible household without one draws one with probability πV . Under Housing First that is the whole rule.

Under Treatment First the programme is conditional. A household fails the condition if it uses, $v > 0$, or has the disorder and is not in treatment, $d = 2$ and $\tau = 0$. A household with a SUD must be in treatment (paid for by the government), as the executive order directs, and no household may use even while in treatment, as the guidance implementing the order requires of drug-free housing.22

Income-eligible holders and income-eligible households that draw a voucher are tested, at a cost ctest each to the government, and one that fails the condition is detected with probability pdet. This means that enforcement is not perfect: with probability $1 - \text{pdet} > 0$ a household that fails the condition actually gets to keep its voucher. A detected holder keeps this period's payment and loses the voucher from the next period; a detected applicant never receives it. In both cases, the household is excluded from the voucher programme for a period of time ($\eta = 2$): it cannot draw a voucher whatever its income, and it returns to the lottery ($\eta = 0$) with probability ψx each year.

None of pdet, ψx and ctest binds in the benchmark, where no voucher is conditional, so all three are set with the experiment in Section 6 rather than in the calibration.

22Executive Order 14321, Section 5(b); the guidance is Addressing Homelessness and Addiction through "Treatment First" (2026), p. 9, issued by the Office of National Drug Control Policy, the Department of Health and Human Services and HUD. The Housing Choice Voucher programme bars admission for current illegal drug use (Code of Federal Regulations (CFR), Title 24, Section 982.553) but does not test; Housing First here is a programme that does not test. The order also encourages the civil commitment of people with mental illness who pose a risk to themselves or others, or who live on the street and cannot care for themselves. In the model, treatment is always the household's choice: a household with a SUD that declines it loses its voucher if the screen catches it.

4.10 The household's dynamic problem

Write Γ

$$\text{[illegible]}\ s' \mid s, h, v, \tau, a'\text{[illegible]}$$

for the law of motion of the state that the transitions above define, given the state and the choices. Then, for $j = 1, \ldots, J$ and with $V(J+1, \cdot) = 0$,

$$V(j, s) = \max \{h, v, \tau, a'\}$$

$$\text{[illegible]}$$

$$U(XP) + \omega c(\xi(j) - 1)\, U(Xc) + \text{א} - \tau\, \bar{\tau}\, u$$

$$+ \beta\, \phi(j, h, d, m)$$

$$\text{[illegible]}$$

S

V (j + 1, s′) Γ

⎮ ds′ | s, h, v, τ, a′⎮

− β χ(j) ωc D(y, d)

⎮

, χ(j) =

⎧ | ⎨

| ⎩

1 − ϕ(j, h, d, m) j < JK 1 j = JK 0 j > JK,

(24)

subject to the budget (16), the income and transfer rules (13) to (15) in Section 4.3, and the restrictions that treatment is available only to a household with d = 2 and that no household begins using after JE.

The first line is the flow of (12). The second is the household's own continuation. The third is the impact of the parent on the child's adult value, counted only when the child leaves: at JK with certainty, and before then with the probability that the parent dies first. D(y, d) is what the parent's disorder costs the child, in the child's own lifetime value,

D(y, d) = E

⎮ V (1, schild)

⎮⎮y, 0 ⎮ − E

⎮ V (1, schild)

⎮⎮y, d ⎮ ≥ 0, (25)

where the child is valued as it enters at j = 1 with the initial states described in Section 4.13, and its productivity is drawn as in (22) in Section 4.8 from a parent with productivity y. In the first term of (25) the parent has never had the disorder, and in the second it has its actual d in the period the child leaves. If it never had a SUD, there is no charge to the parent (D(y, 0) = 0); if it had a SUD before the child leaves (d ≥ 1), it is charged (D(y, d) > 0).23

One weight ωc multiplies both child terms, the flow of childhood and the value of the child's adulthood. At ωc = 0 the damage a household does to its child is entirely external; at ωc > 0 it internalises the share ωc and only 1 − ωc remains external.

The policy functions h(j, s), v(j, s), τ(j, s) and a′(j, s) are the dwelling, drug use, treatment and saving that solve (24) at age j and state s.

4.11 The government

Write Φ(j, s) for the distribution of the households of age j over the states s; its law of motion is in Section 4.13. The government's budget constraint is

R =

J ∫

j=1

∫

S

∫ ˜y(j, y, e) −min

∫ ˜y(j, y, e), τℓ˜y(j, y, e)φ∫

+ τk r a

+

∫ 1 −ϕ ∫ j, h(j, s), d, m

∫∫ a′(j, s)

∫ dΦ(j, s) −T0

−

J ∫

j=1

∫

S

∫ UI(j, y, e) + S(j, y) + F(j, s) + Vch(j, s) + ctest 1test(j, s) + ctreat 1{d = 2, τ(j, s) = 1}

∫ dΦ(j, s).

(26) The first line is revenue from the labour and capital income taxes of (14). The second is the estates of the households who die, which the government collects, less the endowment T0 it gives the unit mass of

23The term is a difference and not a level. Written as a level, ωc times the child's value, the term would be worth more to a household the likelier the household is to die, because the death branch carries it. It would then pay most to the households with a SUD, for reasons that have nothing to do with the child. As a difference, the term is zero for a

parent who has never had the disorder and a cost otherwise.

entrants. The third is what the government pays: UI, UI(j, y, e) = ˜y(j, y, e) 1{j < JR, e = 2}; social security, S(j, y) = S(y) 1{j ≥JR}; food assistance F(j, s) and the voucher Vch(j, s) of (15), at the household's dwelling h(j, s); under Treatment First, the cost ctest of each household tested, 1test(j, s) = 1; and, under both policies, the cost ctreat of a year of treatment for each household with the SUD that takes it up, τ(j, s) = 1, since treatment is publicly funded.

In the benchmark the tax parameters are set as in Section 5 and R is whatever surplus they produce. In every experiment the taxes are held at their benchmark values, and the probability of drawing a voucher πV adjusts so that R equals its benchmark value ¯R, unless the experiment fixes πV .

4.12 The drug market

The price of the drug clears its market,

pv = ν1S1/εs

v , Sv =

J ∑

j=1

∫ v(j, s) dΦ(j, s), (27)

where εs is the price elasticity of supply and ν1 is normalised so that the price is one in the benchmark, and then held fixed across experiments.24

4.13 The law of motion

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

Φ(j+1, S′) =

∫

S

ϕ

∫ j, h(j, s), d, m

∫ Γ

∫ S′ | s, h(j, s), v(j, s), τ(j, s), a′(j, s)

∫ dΦ(j, s), (28)

for j = 1, . . . , J −1 and every measurable set S′ of the state space S of (9), with the integral over the state s of the households of age j, together with the inflow of entrants

Φ(1, S′) =

�

j≤JK

�

Sχ(j) N

� S′ | s

� dΦ(j, s) �

j≤JK

�

Sχ(j) dΦ(j, s) , (29)

where the integrals are again over s, N places the child according to (22) and χ(j) is the weight defined in Section 4.10. An entrant starts with the endowment T0, is employed, has never had the disorder, holds no voucher, and carries SMI with the prevalence observed at age 26.25

4.14 Competitive equilibrium

A competitive equilibrium is a collection of

• value functions

• {Vt(j, ·)}J

j=1

•∞ t=0; • policy functions

• {ht(j, ·), vt(j, ·), τt(j, ·), a′

t(j, ·)}J

j=1

•∞ t=0; • distributions

• {Φt(j, ·)}J

j=1

•∞ t=0; and • prices, policies and aggregates

 pvt, πV t

∞ t = 0, together with ¯y, ymed and FMR, which carry no date because they are kept fixed at their benchmark values,

such that, at every date t,

1. the policy functions solve (24) given its terminal condition, and {Vt(j, ·)}J j = 1 are the associated values;

24The suppliers are outside the economy: what they earn along the supply curve is counted neither in welfare nor in the budget.

25A disorder acquired before that age is therefore acquired in the model only after entry; however, the share of adults with a SUD still matches the data (Section 5).

2. Φt + 1(j + 1, ·) follows from Φt(j, ·) by the law of motion (28) together with the entrants of (29) and (22); 3. the government budget (26) holds at πV t with R = ¯R; 4. the drug market clears at pvt, so (27) holds.

A stationary equilibrium is the special case in which all of these are constant and Φt + 1 = Φt, which is what the calibration and the long-run policy comparisons report. An experiment that holds πV or the drug price at its benchmark value drops the budget or the drug-market condition. The sequence version is what the transition between two stationary equilibria solves.

5 Calibrating the quantitative model

The model has fifty-nine parameters, some of which vary by age. Thirty-eight are calibrated externally, from data or the literature, and twenty-one internally. The twenty-six that carry this paper's story are in Table 3, with the moment, source, data and model value behind each, and are discussed below in numbered blocks of parameters that work through similar mechanisms. The other thirty-three, common to most life- cycle models, are in Appendix A to save the reader time; its blocks continue the numbering. Three further parameters govern drug testing and exclusion from the voucher; they play no role in the benchmark and are set with the experiment in Section 6. data/DATA APPENDIX.md in the replication package documents the dataset, year, universe and computation behind every estimate, and the package carries all datasets, so nothing has to be downloaded.

5.1 The twenty-six parameters that carry the story of this paper

Table 3: The twenty-six parameters that carry the story of this paper

Parameter Description Set Moment and source Data Model Value

Block 1: Who uses drugs θ Weight on drugs in utility Int. Drug spending over income, Midgette et al. (2019) 16.0% 14.8% 0.165 b Marginal utility from the first unit of drugs Int. Any drug use disorder, NSDUH 2024 8.90% 8.96% 0.0054 πd Chance that a year of use leads to a SUD Int. Share of users with a SUD, NSDUH 2024 41.3% 42.6% 0.0345 κφd Increased mortality with a SUD Int. Income of drug users over average, NSDUH 2024 74% 62% 4.00 א Flow value of being alive Int. VSL, Viscusi and Aldy (2003) 80 81 87.6

Block 2: Costs of homelessness and vouchers ʁue